

Why 50% Dot-Plane Swaps Cannot Cleanly Decouple Depth × Color

Given: S0 (coherent translator) fixed at Near=Red throughout · Rule: single color per depth plane at all times

Starting configuration (Cond 1, no swap)

Subfield	Depth	Color	Role	
S0 ●	Near	Red	Coherent translator — FIXED throughout	◀ cannot change
S1 □	Near	Red	Noise companion of translator	
S2 ▲	Far	Green	Coherent rotator	
S3 ◇	Far	Green	Noise rotator	

The constraint chain

1.

S0 pinned at Near+Red
Near plane must be Red (S0 is there and is Red).
2.

Any dot in Near must be Red
Moving any dot to Near without making it Red → mixed colors in Near plane. ✗
3.

Any dot in Far must be Green
Far is the only other plane; it must be a single color. Green is the only option.
4.

Depth change = forced color change
Moving a dot between planes forces it to adopt the destination plane's color.
You cannot independently manipulate depth without also changing color.

All possible partial swaps — outcomes

Subfields changed	Change type	Near plane	Far plane	Clean planes?	Problem
S1 only	depth only (Near→Far)	S0(R)	S1(R)+S2(G)+S3(G)	✗	Far: mixed R+G
S1 only	color only (Red→Grn)	S0(R)+S1(G)	S2(G)+S3(G)	✗	Near: mixed R+G
S2 only	depth only (Far→Near)	S0(R)+S1(R)+S2(G)	S3(G)	✗	Near: mixed R+G
S2 only	color only (Grn→Red)	S0(R)+S1(R)	S2(R)+S3(G)	✗	Far: mixed R+G
S3 only	depth only (Far→Near)	S0(R)+S1(R)+S3(G)	S2(G)	✗	Near: mixed R+G
S3 only	color only (Grn→Red)	S0(R)+S1(R)	S2(G)+S3(R)	✗	Far: mixed R+G
S1+S2	depth+color	S0(R)+S2(R)	S1(G)+S3(G)	✓*	S2 leaves Far; Near has S0+S2 → density change, S0 isolated
S1+S3	any	S0(R)+S1(?) or S3(?)	—	✗	Cannot recolor without mixing planes
S2+S3	color only (Grn→Red)	S0(R)+S1(R)	S2(R)+S3(R)	✓*	Both planes Red — color distinction lost
S2+S3	depth only (Far→Near)	S0+S1+S2+S3 (all Near!)	—	✗	Far plane empty; Near: mixed R+G
S2+S3	depth+color (Far→Near+Red)	all 4 subs (R)	—	✗	Far plane empty; no depth distinction
S1↔S3 (swap planes)	depth+color both change	S0(R)+S3(R)	S1(G)+S2(G)	✓*	Clean, but depth+color tied together — not independent

Conclusion

- No viable 50% swap independently manipulates depth OR color while keeping S0 fixed and both depth planes occupied with a single color each.
- The three ✓* rows above either lose the color distinction (both planes become the same color), empty one plane, or bundle depth and color changes together (S1↔S3 swap) — preventing independent decoupling.

→ Therefore: 100% whole-field swaps (both fields change together) are the minimal condition for clean independent Depth × Color manipulation. This is why Conds 1-8 are designed the way they are.